**Saleh Aldhafeeri**
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## EDUCATION & CERTIFICATIONS

**The Pennsylvania State University** **University Park, Pennsylvania**
*B.S in Aerospace Engineering* *Jan 2023 - May 2026*

- **GPA:** 3.65/4

**Certifications:** Ansys STK Level 3 Grand Master, Ansys Fluent Certification, Solidworks CSWE, MathWorks Certified MATLAB Associate

## SKILLS

**Software:** ANSYS Fluent, SolidWorks CAD design, SolidWorks Flow Simulation, MATLAB, Python, JavaScript, SQL, AMOS.
**Technical Areas:** Systems engineering, GNC, EKF, inertial navigation, spacecraft mission architecture, compressible CFD, airworthiness compliance

## EXPERIENCE

**Saudia Technic.** **Jeddah, Saudi Arabia**
Aircraft Engineering Intern. *May 2025 – Sept 2025*

- Closed technician raised maintenance queries by diagnosing reported aircraft issues, selecting compliant action from CMM, AMM, and SRM manuals then answering with engineering orders for approval.
- Moved maintenance cases from open query to approved execution by presenting technically justified repair paths in daily reviews with my supervising engineer.
- Safeguarded post maintenance airworthiness by verifying completed work against Boeing & Airbus requirements, and confirming maintenance related weight and balance changes remained within allowable aircraft limits before release approval.

## PROJECTS

**Project Manager for Prometheus Mission I Satellite Capstone Project.** *Aug 2025 – Dec 2025*

- Led a 6-engineer team delivering full design artifacts for an autonomous on-orbit refueling satellite mission across 4 mission phases: rendezvous, docking, propellant transfer, and departure.
- Directed cross-functional integration across propulsion, GNC, thermal, and structures, delivering a mass- and power-constrained design with autonomous fault response..

**Extended Kalman Filter for Lunar Navigation** *Jan 2025 – May 2025*

- Designed a 9-state EKF in MATLAB for lunar navigation with 100 Hz propagation and multi-rate IMU fusion; validated estimator consistency within ±2σ bounds..

**Autonomous Drone Landing on Moving object -GNC System** *Jan 2025 – May 2025*

- Designed a cascaded PID/PD quadrotor guidance and control system for autonomous landing on a moving vehicle; achieved centimeter-level touchdown accuracy via velocity feedforward and flight-phase logic.

**Aerodynamic Turbine Blade Design and CFD.** *Aug 2024 – Dec 2024*

- Modeled a parametric 3D turbine blade in SolidWorks with adjustable chord, camber, and twist; reduced total-pressure loss by 12% and raised section L/D by 18% via CFD optimization.

## LEADERSHIP & ACTIVITIES

**Treasurer at Penn State MENA Caucus Club.** *Dec 2023 – Dec 2024*

- As Treasurer, built and launched a secure finance website for 100+ members using JavaScript for frontend, Python for backend, and PostgreSQL while managing a $10K event budget and initiating 7 fundraisers.

**Nittany Lion Consulting Group Member.** *Jan 2023 – Dec 2025*

- Consulting university club focused on structured problem solving, market research, financial modeling, and hypothesis driven case analysis.